# Insights on the statistics of extreme Arctic sea ice conditions from the EC-Earth3 climate model

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## 1. Objective

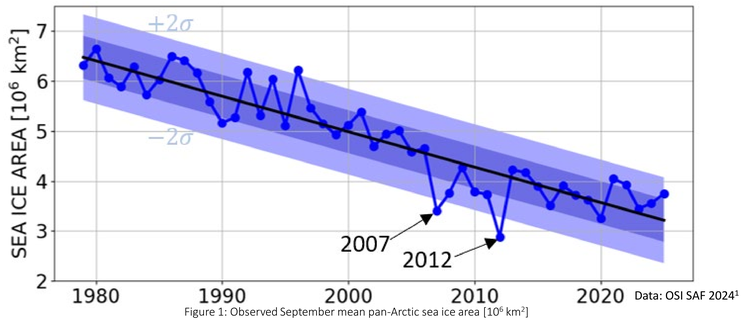



Figure 1: Observed September mean pan-Arctic sea ice area [$10^6$ km$^2$]

Goals: What are systematic precursors of extremes of sea ice reduction considering a robust statistics on these events?
How likely (in a statistical sense) is a 2012 sea ice loss event for a given initial condition and/or climate state?

Problem: Quantitative statistical and dynamical studies of climate extremes hindered by lack of data
→ From statistical physics: improve the sampling efficiency of extreme events with rare event algorithms
→ Genealogical selection algorithm adapted from Del Moral and Garnier (2005)[3]; Giardina et al. (2011)[4] (Ragone et al. 2018[5]; Ragone and Bouchet 2019[6]; 2021[7], Sauer et al. 2024[8]; 2025[9]): Efficient to study time-persistent extremes

## 2. Methodology: Rare event algorithm

- Importance sampling of trajectories in ensemble simulation with numerical model
  → Make trajectories leading to large anomalies of a time-averaged observable common
  → Better statistics on extreme events (composites, return times) + generation of ultra-rare but high impact events
- Resampling at constant time intervals: trajectories are killed or cloned depending on weights quantifying the time-average of the target observable during a past interval
- Reconstruction of effective ensemble based on surviving trajectories
- Importance sampling formula: relates probabilities of trajectories between biased and unbiased statistics

$$P_k\left(\{X(t)\}_{0\le t\le T_a}\right) = \frac{e^{k\int_0^{T_a} A(\{X(t)\})\,dt}}{Z_K} P_0\left(\{X(t)\}_{0\le t\le T_a}\right)$$

$P_k, P_0$: prob. dens. in biased and unbiased statistics
$k, Z_k$: biasing parameter and normalization term
$t, T_a$: time and simulation length
$A, \{X_n(t)\}$: observable and model trajectories

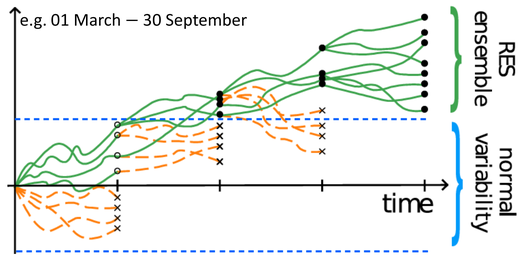



Figure 2: Illustration of the rare event algorithm.
Figure from Wouters et al. 2023[10]

## 3. Model and experimental design

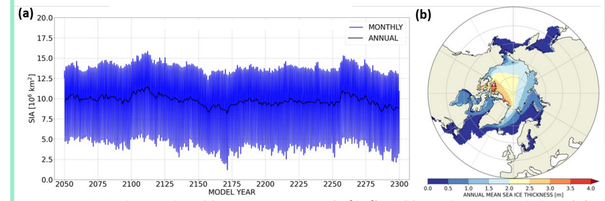



Figure 3: EC-Earth3 control run: (a) Pan-Arctic sea ice area [$10^6$ km$^2$] and (b) annual mean sea ice thickness [m]

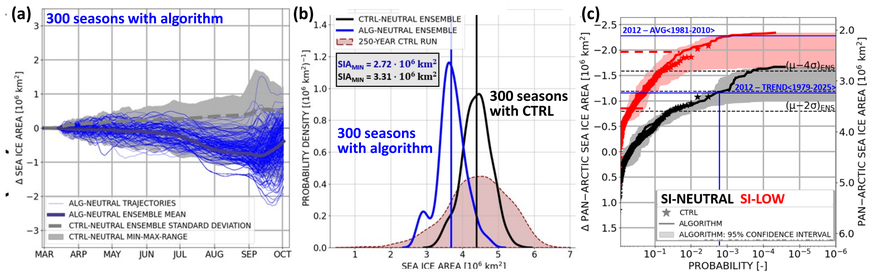
EC-Earth3.3.1: CMIP6-generation European community Earth-System-Model verion 3 (Döscher et al. 2022[11])

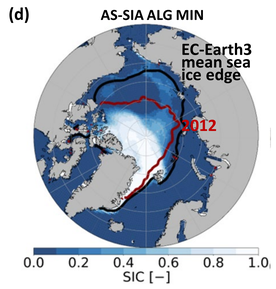
Components and resolutions: IFS-36r4: T255L91, NEMO3.6-LIM3: ORCA1L75

Stationary forcing: year-2000 greenhouse gas conditions and solar forcing

Rare event ensemble simulations in seasonal climate prediction setup:
→ N=300 trajectories running from 01 March to 30 September
→ Two experiments with two different sea ice-ocean initial conditions:
(A): "SI-NEUTRAL":
initial sea ice thickness representative of the 1979-2015 climatological mean of PIOMAS (Zhang and Rothrock[12])
(B) "SI-LOW": as (A), but after adding the difference of gridded snow-sea ice-ocean variables between 2012 and the 1995-2005 mean using "anomaly initialization" (Tian et al. 2021[13])

## 4. Importance sampling of low sea ice states in EC-Earth3




- Importance sampling of trajectories leading to low late summer sea ice area
- Simulation of events with probabilities of less than $10^{-4}$ per year with a computational cost of order $10^2$ years
- Spatial distribution of sea ice cover in the most extreme trajectory available from the experiments physically compatible with the 2012 event

Figure 4: (a,b,d) SI-NEUTRAL: (a) Daily sea ice area anomalies relative to the control ensemble mean, but obtained with the algorithm. (b) PDFs of August-September (AUGSEP) mean sea ice area for (blue) the algorihm, (black) the control ensemble and (shading) the 250-year control run. (d) AUGSEP mean sea ice concentration in the most extreme sea ice low available from the ensemble compared to the sea ice edges (red) observed in September 2012 (OSI SAF 2017[2]) and (black) in the control ensemble mean. (c) SI-NEUTRAL and SI-LOW experiments for (stars) the control ensembles and (solid lines) the rare event algorithm. The curves show the estimated probabilities of occurrence (x-axes) of AUGSEP mean sea ice area values below a given value (y-axes). Shading denotes 95% confidence intervals.

## 5. Statistical link between extreme sea ice reduction and the atmospheric circulation

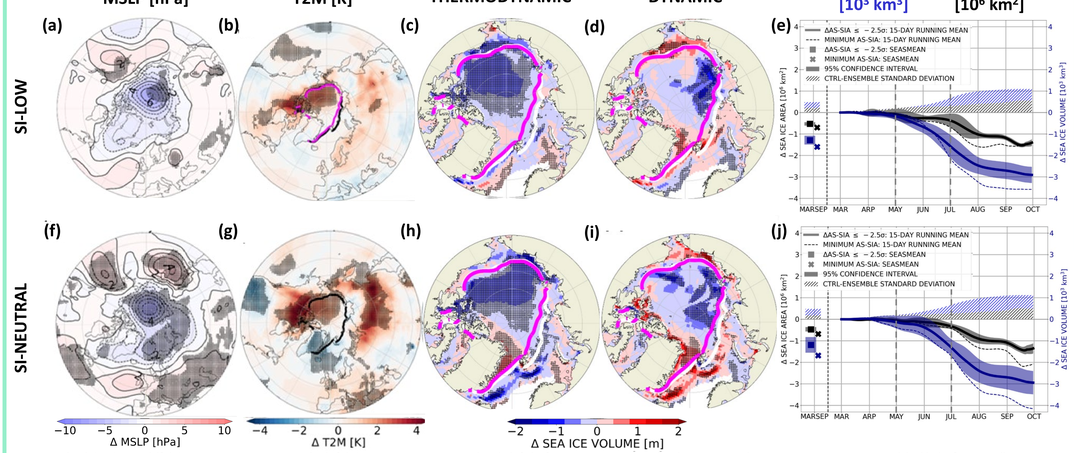



Figure 5: (a-e) SI-LOW and (f-j) SI-NEUTRAL experiments: (a-d, f-i) Composite mean anomalies (a,b,f,g) of May-June mean sea level pressure and two metre temperatures and (c,d,h,i) of seasonally-integrated thermodynamic and dynamic sea ice volume change over all trajectories with an August-September mean sea ice area of at least 2.5 control ensemble standard deviations below the control ensemble mean. Hatching denotes statistical significance at the 5% level. (e,j) Seasonal evolution of pan-Arctic (blue) sea ice volume and (black) sea ice area anomalies relative to the control ensemble mean. Thick lines are composites as in (a-d, f-i) and shading denote 95% confidence intervals. Thin lines correspond to the most extreme sea ice low trajectory available from the ensembles respectively.

## 6. Conclusions

- Using a statistical physics-based rare event algorithm, we perform for a first time a quantitative statistical analysis of the precursors of extremes of Arctic sea ice reduction in a comprehensive climate model.
- While preserving the dynamical consistency of the model, the algorithm allows to compute the probabilities of extreme fluctuations such as in 2012 → $P(2012)_{\text{EC-Earth3}} \approx O(10^{-2}\text{ - }10^{-3})$ in a present-day climate
- On statistical average, extreme late summer sea ice lows are systematically preceded by enhanced sea ice volume loss from May to July, implying a substantial increase of predictability during that period.
- Enhanced May-July sea ice volume loss is systematically associated with negative mean sea level pressure anomalies over the central to eastern Arctic Ocean. Cyclonic flow favors thermodynamic (dynamic) sea ice loss on the North American-Pacific (Eurasian) side of the Arctic Ocean.
- In the future, the algorithm could be useful for attribution and impact studies. Likewise, it could help to improve seasonal predictions of summer sea ice conditions using data-based statistical models, machine learning-based forecasts and climate model ensemble-based initialized forecasts.

This communication is supported by the French Community of Belgium as part of a FRIA grant.

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